

上册部分

一、填空题 (每空 0.5 分, 共 10 分)

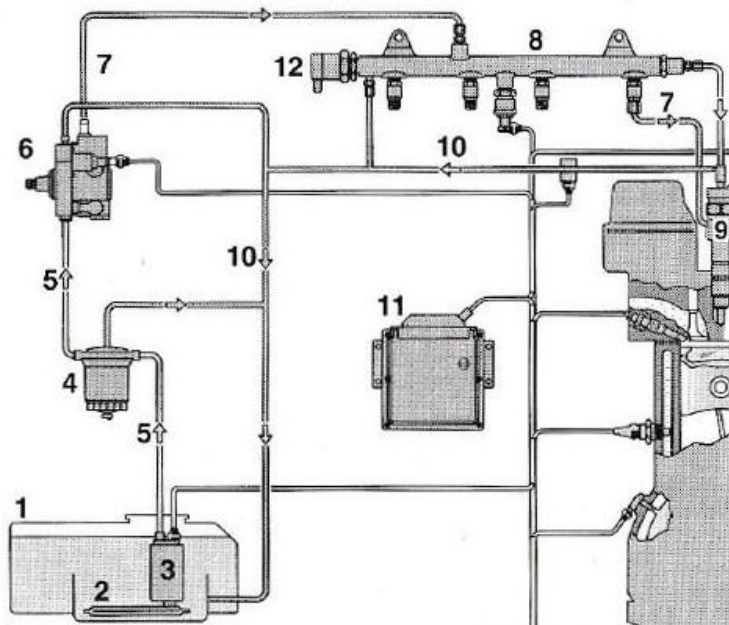
- 1) The components of a piston gasoline engine are basically divided into engine block, crank and connecting rod mechanism, valve and valvetrain mechanism, ____, ____, cooling system, ____ and starting system.
- 2) The working cycle of a four-stroke internal combustion engine is sequentially consisted of ____, ____, expansion stroke and ____.
- 3) When the piston is at ____, it is at the farthest position from the crankshaft axis. On the contrary, the piston is at ____ where it is at the closest position with the crankshaft axis.
- 4) 活塞式内燃机机体组主要包括气缸盖罩、____、____、____、油底壳等。
- 5) 活塞式内燃机燃烧室是由缸套内壁、____、____和 气门底面等围成的空间。
- 6) 车用废气涡轮增压器由____、中间体、____三部分组成。
- 7) 配气机构的凸轮轴转速是曲轴转速的____。凸轮轴与曲轴之间一般采用____传动、____传动、齿形带传动等传动方式。
- 8) 曲轴主轴颈处的润滑方式为____, 活塞与缸套之间的润滑方式为_____。

二、计算、概念和简答题 (每小题 4 分, 共 20 分)

- 1) There is a 3-cylinder gasoline engine that its compression ratio is 11, its cylinder diameter is 80mm, and its stroke is 100mm. Please calculate the working volume (unit: mL), combustion chamber volume (unit: mL) of its single cylinder and the displacement of the engine (unit: L). *Note: Don't use calculator. Keep two decimals (小数点后两位) in the results.*
- 2) There is a gasoline engine with intake valve opening (α) at 15°CA BTDC and closing (β) at 30°CA ABDC , exhaust valve opening (γ) at 40°CA BBDC and closing (δ) at 10°CA ATDC . Please draw its valve timing diagram and mark $\alpha, \beta, \gamma, \delta$. And calculate the valve overlap, intake and exhaust duration in crank angle.
- 3) 请介绍现代直喷汽油机常用的排气后处理装置及其基本功能。
- 4) 请说明强制曲轴箱通风系统的基本工作过程和控制方法。
- 5) 请介绍目前车用质子交换膜单体燃料电池的基本组成和功能。

三、论述题 (每小题 10 分, 共 20 分)

- 1) With reference to the figure at left bottom, please list the main components and their fundamental functions for the crank and connecting rod mechanism, and the valve and valvetrain mechanism.
- 2) 参考右下图, 写出现代电控高压共轨柴油机燃油供给系统的组成、控制方法和工作特点。



- 一、
 - 1) 供给系统、润滑系统、点火系统(fuel and air supplying system, lubrication system, ignition system)
 - 2) 排气冲程、压缩冲程、进气冲程(exhaust stroke, compression stroke, intake stroking)
 - 3) 上止点、下止点(TDC, BDC)
 - 4) 气缸盖、气缸垫、气缸体
 - 5) 活塞顶, 缸盖底面
 - 6) 涡轮机, 压气机
 - 7) 1/2, 齿轮、链
 - 8) 压力润滑、飞溅润滑

二、

1)

工作容积(working volume): $3.14/4 \times 8 \times 8 \times 10 = 502.40 \text{ mL}$;

燃烧室容积(combustion chamber volume): $502.4 / (11-1) = 50.20 \text{ mL}$;

排量(displacement): $3 \times 0.5024 = 1.51 \text{ L}$

2)

配气相位图(valve timing diagram): 参考右图。

气门重叠角(valve overlap): $\alpha + \delta = 15 + 10 = 25^\circ \text{ CA}$

进气持续角(intake duration): $\alpha + 180 + \beta = 15 + 180 + 30 = 225^\circ \text{ CA}$

排气持续角(exhaust duration): $\gamma + 180 + \delta = 40 + 180 + 10 = 215^\circ \text{ CA}$

3)

TWC—氧化 HC、CO, 还原 NO_x

GPF—过滤颗粒物

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4)

新鲜空气从外界进入曲轴箱, 携带挥发燃油/润滑油/水等, 进入进气歧管, 再吸入缸内燃烧。

主要提过 VOC 阀控制

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5)

正极板: 导电, 提供氢气流场; 气体扩散层: 使气体分布均匀; 膜电极: 催化剂层, 质子交换膜; 负

极板: 导电, 提供空气流场/水流道

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三、

1)

曲柄连杆机构:

活塞组—活塞、活塞环(气环、油环)、活塞销、卡环

连杆组—连杆小头衬套、连杆(小头、杆身、大头)、连杆大头端盖及连接螺栓、连杆轴瓦

曲轴飞轮组—曲轴、飞轮、扭转减振器、主轴

配气机构:

气门组—气门、气门弹簧、气门弹簧座、气门锁夹

气门传动组—正时齿轮、链条/齿形带/齿轮、凸轮轴, 挺柱, 摇臂, 推杆

2)

燃油供给: 油箱、低压油泵、油滤、高压油泵、供给管路、油轨、喷油器、回油管路;

控制方法: 根据负荷/转速等工况参数电控调节喷油脉宽、喷油次数等; 轨压闭环控制;

工作特点: 燃油工作压力非常高, 控制灵活多变

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